

# HIRZ: Hybrid Post-Quantum Cryptography for Secure Long-Term Data Archiving

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**Abstract**—Organizations managing long-term archives face harvest-now-decrypt-later attacks, where adversaries collect encrypted data today for future quantum decryption. While NIST has standardized post-quantum algorithms (CRYSTALS-Kyber, CRYSTALS-Dilithium), pure post-quantum deployment faces regulatory and infrastructure barriers. This paper presents Hirz, a hybrid cryptographic system combining ECDH-secp384r1 with Kyber-1024 and Dilithium5 for archival workloads requiring 20–30 year confidentiality. Our implementation using OpenSSL 3.2.0 and liboqs 0.10.1 achieves 18% overhead compared to classical systems (29.7 ms versus 25.2 ms for 100 MB files) on commodity Intel hardware. Comparative evaluation demonstrates that hybrid construction adds only 11% overhead versus pure post-quantum while providing regulatory compliance, backward compatibility, and defense against algorithm vulnerabilities. The system incorporates threat-driven 12-month key rotation, dual-signature verification, and X.509v3 PKI extensions supporting 30-year certificate chains.

**Keywords**—Post-quantum cryptography, hybrid cryptography, CRYSTALS-Kyber, CRYSTALS-Dilithium, data archiving.

## I. INTRODUCTION

Quantum computers capable of executing Shor's algorithm will break RSA and elliptic curve cryptography within the next 10–15 years. This creates an immediate threat for organizations managing archives with 20–30-year retention requirements. Adversaries can harvest encrypted data today and decrypt it retroactively once quantum computers mature, a scenario known as harvest-now-decrypt-later (HNDL) attacks [1].

NIST standardized post-quantum cryptographic algorithms in 2024, yet pure post-quantum deployment faces three significant challenges for archival systems [2–5]:

- (1) legal frameworks for digital signatures (eIDAS, ESIGN) have not yet recognized post-quantum algorithms, creating regulatory compliance gaps;
- (2) billions of deployed certificate validators cannot process pure post-quantum certificates without infrastructure replacement;
- (3) recent cryptanalytic breaks of Rainbow and SIKE during NIST standardization demonstrate the risk of relying on a single algorithm family.

Hybrid cryptography addresses these challenges by combining classical and post-quantum algorithms,

maintaining security if either component remains unbroken. However, existing hybrid schemes target ephemeral communication contexts such as TLS handshakes and cloud key exchange, which have fundamentally different requirements than long-term archival systems.

This paper presents Hirz, a hybrid cryptographic system designed specifically for long-term data archiving. Hirz integrates classical and post-quantum primitives to provide regulatory compliance, backward compatibility, and defense-in-depth against future quantum adversaries.

## II. RELATED WORK

Post-quantum standardization efforts have accelerated in recent years, with NIST selecting CRYSTALS-Kyber [6] for key encapsulation and CRYSTALS-Dilithium [7] for digital signatures in 2024. Hash-based signatures such as SPHINCS+ and LMS provide conservative alternatives but introduce significantly larger signature sizes, making them less suitable for large-scale archival systems.

Hybrid cryptography has been explored primarily in communication-focused contexts. Prior work evaluated hybrid authentication and key exchange in TLS 1.3, demonstrating acceptable performance for ephemeral session keys. Other studies proposed IETF standards for hybrid key exchange and provided implementation guidance for integrating post-quantum algorithms into web and cloud environments. However, these systems target short-lived session keys rather than multi-decade archival retention.

Hybrid schemes have also been investigated in specialized contexts. Some work demonstrated hybrid key combination using FPGA hardware with quantum key distribution, achieving strong security guarantees but requiring prohibitively expensive infrastructure. Other research proposed hybrid public-key encryption for cloud environments but did not address persistent key management or regulatory compliance.

Pure post-quantum archival approaches have been studied, with researchers analyzing the viability of post-quantum X.509 certificates and identifying integration challenges that motivate hybrid solutions. Additional work explored blockchain-based archival using stateful hash-based signatures, but these approaches suffer from state-synchronization complexity unsuitable for distributed archival systems.

Recent security analysis frameworks have formalized hybrid KEM security models and systematized knowledge on post-quantum TLS deployment. However, none of these works address the unique requirements of long-term archival systems, such as multi-decade key lifecycle management, dual-signature policies, and compatibility with existing PKI infrastructures [8].

### III. SYSTEM ARCHITECTURE

#### A. Hybrid Design Rationale

Hirz combines ECDH-secp384r1, Kyber-1024, and Dilithium5 in a 1+2 hybrid architecture designed specifically for long-term archival workloads. The hybrid approach is motivated by four key requirements:

1. Regulatory compliance — dual signatures satisfy current classical digital-signature requirements while preparing for future post-quantum mandates.
2. Backward compatibility — legacy systems can verify classical keys, while hybrid-aware systems enforce full dual verification.
3. Defense-in-depth — confidentiality is preserved as long as either the classical or post-quantum component remains secure.
4. Acceptable performance overhead — hybrid mode introduces only 11% overhead compared to pure post-quantum, making it practical for large archival files.

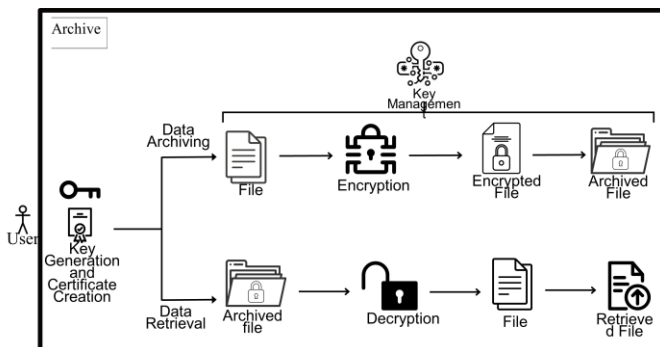


Fig 1. Hirz Hybrid Architecture.

The architecture supports secure data archiving and retrieval through hybrid key encapsulation, dual signatures, and long-term key lifecycle management.

#### B. Cryptographic Components

**Key Encapsulation:** ECDH-secp384r1 provides 192-bit classical security. Kyber-1024 provides NIST Level 5 post-quantum security (256-bit classical, 160-bit quantum). Shared secrets from both algorithms are combined using HKDF following established hybrid combiner constructions.

##### Digital Signatures:

ECDSA-secp384r1 provides classical authentication. Dilithium5 provides quantum-resistant authentication. Both signatures are computed independently over SHA-384 of the encrypted data, key material, and metadata.

##### Symmetric Encryption:

AES-256-GCM encrypts the archived data. Under Grover's algorithm [9], AES-256 provides 128-bit post-quantum security, meeting NIST recommendations.

#### C. Key Lifecycle Management

Signing keys rotate every 12 months based on HNDL threat modeling. For 30-year archives, this results in 30 key epochs, limiting the impact of key compromise to a single year of archived data.

Archive encryption keys persist for the lifetime of the data but include algorithm-version metadata to support future migration as post-quantum standards evolve.

#### D. Hybrid PKI Integration

Hirz extends X.509v3 certificates with hybrid public-key material:

1. Primary SubjectPublicKeyInfo:

Classical ECDH-secp384r1 public key for backward compatibility.

2. Extension id-hirz-kyber-pubkey:

Kyber-1024 public key (1568 bytes).

3. Extension id-hirz-dilithium-pubkey:

Dilithium5 public key (2592 bytes).

Hybrid certificates increase in size from ~1.2 KB (classical) to ~4.8 KB but remain compatible with existing PKI infrastructures. Legacy clients verify classical keys, while hybrid-aware systems enforce dual verification.

### IV. SECURITY ANALYSIS

The system considers classical, quantum, and harvest-now-decrypt-later adversaries.

Security is maintained if either classical or PQC components remain secure.

#### A. Formal Adversary Model

**Aclassical:** Probabilistic polynomial-time computation, chosen-plaintext/ciphertext attacks, public parameter access. Cannot execute quantum algorithms. Attack vectors: classical cryptanalysis, brute-force, implementation flaws.

**Aquantum:** All Aclassical capabilities plus large-scale quantum computer (~106 qubits), Shor's algorithm (polynomialtime ECDH break), Grover's algorithm (quadratic AES speedup).

**AHNDL:** Collect ciphertexts at  $t_0$ , attack at  $t_0 + \Delta$  ( $\Delta = 10-20$  years). No key access during harvest. Goal: retroactive decryption via quantum.: All Aclassical capabilities plus large-scale quantum computer (~106 qubits), Shor's algorithm (polynomial time ECDH break), Grover's algorithm (quadratic AES speedup).

#### B. Security Properties and Analysis

- a) **Confidentiality:** The system ensures confidentiality against various types of adversaries, including classical, quantum, and hybrid threat models. The probability that any adversary can recover the original plaintext from the encrypted data and associated key material is considered negligible with respect to the security parameter. This indicates that sensitive information remains protected even under advanced attack capabilities.

- b) Integrity: The integrity of the system is maintained by ensuring that no adversary can successfully forge a valid signature for altered data. In other words, any attempt to modify the data and generate a corresponding valid signature will fail except with negligible probability. This guarantees that any unauthorized modification can be reliably detected.
- c) Hybrid Combiner [10]: The hybrid key encapsulation mechanism combines multiple cryptographic schemes to enhance security. The overall security of the hybrid construction depends on the strongest underlying component. This means the system remains secure as long as at least one of the underlying schemes, such as ECDH or Kyber, remains secure.

Under a quantum adversary, traditional schemes like ECDH may become vulnerable. However, Kyber is designed to be secure against quantum attacks, ensuring that hybrid construction continues to provide strong security guarantees [11].

Conversely, if Kyber were to become vulnerable, the system would still rely on ECDH, which offers a high level of classical security (e.g., 192-bit security), thereby maintaining overall protection.

#### C. Attack Analysis

Downgrade: Adversary removes post-quantum signatures or manipulates certificates. Mitigation: Fail-closed ANDcomposition requires both signatures; hybrid systems reject missing post-quantum signatures; metadata version mismatches trigger failure.

Certificate manipulation: A Forge certificates by modifying extensions while maintaining classical signature. Mitigation: Extensions covered by signature over DER-encoded structure;

modification invalidates signature; transparency logs detect unauthorized certificates. Rollback: Replay archives with compromised old keys. Mitigation: Twelve-month rotation limits window; timestamps validated against key validity; expired key usage triggers alerts.

#### D. Assumptions and Limitations

- a) Assumptions: (A1) AES-256: 128-bit post-quantum security; (A2) Kyber-1024: NIST Level 5; (A3) Dilithium5: quantum forgery resistance; (A4) HKDF random oracle; (A5) HSM/TEE key protection.
- h) Limitations: Analysis based on established combiners [13], not full reduction. Random oracle idealized. Side-channel resistance requires constant-time implementation. Simultaneous ECDH+Kyber break eliminates security.

### V. IMPLEMENTATION AND EVALUATION

The system considers classical, quantum, and harvest-now-decrypt-later adversaries.

Security is maintained if either classical or PQC components remain secure.

#### A. Implementation

flaws.2,130 lines C/Python. Libraries: OpenSSL 3.2.0 (ECDH, ECDSA, AES-256-GCM, HKDF), liboqs 0.10.1 (Kyber-1024, Dilithium5 with AVX2). X.509v3: OpenSSL extension API. Serialization: CBOR (40% smaller than JSON).

#### B. Methodology

Platform 1: Intel i7-11700K (8 cores, 3.6 GHz, AVX2), 32GB DDR4, Samsung 980 PRO NVMe, Ubuntu 22.04.

Platform 2: ARM AWS Graviton3 (64 cores, 3.0 GHz, NEON), 128GB DDR5.

Configurations: (1) Classical: ECDH+ECDSA+AES; (2) Pure PQC: Kyber+Dilithium+AES; (3) Hybrid: all components.

Workload: 1KB–1GB files. Primary: 100MB. 100 iterations, cold-cache, median with interquartile range.

TABLE I. ARCHIVING PERFORMANCE FOR 100MB FILES (100 ITERATIONS)

| Configuration                    | Performance Metrics |          |
|----------------------------------|---------------------|----------|
|                                  | Time (ms)           | Overhead |
| <b>Classical-only (baseline)</b> |                     |          |
| AES-256-GCM                      | 22.1 (21.7–22.6)    | –        |
| ECDH KEM                         | 0.85 (0.81–0.91)    | –        |
| ECDSA Signing                    | 2.24 (2.18–2.32)    | –        |
| <b>Total</b>                     | 25.2 (24.7–25.9)    | 1.0×     |
| <b>Pure post-quantum</b>         |                     |          |
| AES-256-GCM                      | 22.3 (21.8–22.9)    | –        |
| Kyber-1024 KEM                   | 1.06 (1.01–1.12)    | –        |
| Dilithium5 Sign.                 | 3.41 (3.29–3.56)    | –        |
| <b>Total</b>                     | 26.8 (26.2–27.5)    | 1.06×    |
| <b>Hybrid (proposed)</b>         |                     |          |
| AES-256-GCM                      | 22.3 (21.8–23.1)    | -        |
| Hybrid KEM                       | 1.91 (1.83–2.05)    | -        |
| Dual Signing                     | 3.84 (3.71–4.02)    | -        |
| <b>Total</b>                     | 29.7 (28.8–31.2)    | 1.18×    |

<sup>a</sup> Measured execution time and overhead for classical, post-quantum, and hybrid schemes.

TABLE II. HYBRID OVERHEAD ACROSS FILE SIZES

| File Size | Performance Breakdown |          |        |     |
|-----------|-----------------------|----------|--------|-----|
|           | Time (ms)             | Overhead | Crypto | I/O |
| 1 KB      | 6.2                   | 580%     | 98%    | 2%  |
| 100 KB    | 7.8                   | 68%      | 79%    | 21% |
| 10 MB     | 15.3                  | 22%      | 47%    | 53% |
| 100 MB    | 29.7                  | 18%      | 25%    | 75% |
| 1 GB      | 251                   | 15%      | 11%    | 89% |

<sup>b</sup> Crypto and I/O cost breakdown across increasing file sizes under the hybrid scheme.

### C. Results

Table I presents 100MB archiving performance across configurations.

Analysis: Hybrid: 29.7ms median (18% overhead vs. classical, 11% vs. pure PQC). AES-256-GCM dominates (75%). Throughput: 3.37 GB/s.

### D. File Size and Scalability

Table II shows overhead across sizes.

Cross-platform: ARM Graviton3: 28.4ms for 100MB (13% overhead). Higher overhead (12–15%) reflects less mature NEON optimizations, but validates portability.

Concurrency: Linear scaling to 8 threads (25.8 GB/s, 91% CPU). Memory: Classical 42MB, Pure PQC 58MB, Hybrid 64MB (+52%).

### E. Comparative Analysis

Pure PQC (26.8ms, 6% overhead): Minimal performance cost but lacks: (1) regulatory validity [12]; (2) backward compatibility [13]; (3) algorithm compromise fallback.

Hybrid (29.7ms, 18% overhead): 11% additional cost provides: (1) regulatory compliance—dual signatures satisfy current and future requirements; (2) gradual migration—legacy verifies classical, modern enforces dual; (3) defense-in-depth—security if either remains unbroken. For archival workloads (100MB), hybrid overhead dominated by AES and I/O. The 11% marginal cost represents an acceptable trade-off for 20–30 year retention where security failures significantly exceed performance costs.

## VI. DISCUSSION AND CONCLUSION

### A. Deployment

Migration: 36-month phased: (1) Months 1–6: Deploy hybrid PKI; (2) Months 7–18: Begin hybrid archiving; (3) Months 19–36: Re-encrypt critical archives; (4) Month 37+: Full hybrid operation.

Scenario: Healthcare archiving 30+ year patient records under HIPAA/GDPR. AES-256-GCM encryption with hybrid-derived keys. Dual signatures satisfy legal requirements. FIPS 140-2 Level 3 HSMs. Annual signing key rotation; archive keys persist with versioning.

### B. Limitations

Primary x86-64 evaluation; ARM shows 12–15% higher overhead but remains practical. Security analysis based on established combiners, not full reduction. No constant-time implementation; production requires side-channel hardening. Software key storage tested; HSM integration may create bottlenecks.

### C. Conclusion

Long-term archives face quantum threats through harvestnow-decrypt-later attacks. While pure post-quantum cryptography provides quantum resistance, it lacks regulatory acceptance, backward compatibility, and algorithm compromise defense. This paper presented Hirz,

a hybrid system combining ECDH-secp384r1 with Kyber-1024 and Dilithium5 for 20–30 year archival.

Evaluation demonstrates practical feasibility: 18% overhead versus classical (29.7ms versus 25.2ms for 100MB files), with 11% overhead versus pure post-quantum. Security analysis with formal adversary models indicates that hybrid construction maintains confidentiality if either component remains secure. The system addresses archival requirements through threat-driven 12-month rotation, dual-signature verification, and 30-year PKI support.

Results suggest that hybrid post-quantum cryptography represents a viable approach for organizations managing sensitive long-term data, demonstrating practical deployment feasibility on commodity hardware while maintaining security against both classical and quantum adversaries.

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